Combitube *Prodigy Anesthesia*

The Combitube is a double-lumen airway device that is blindly placed into the hypopharynx. The distal lumen has a large orifice which can be used to decompress the stomach when the tube is placed into the esophagus. The second lumen has a multiorifice opening that faces the larynx and is used as the airway when placement is in the esophagus. The combitube has an esophageal cuff and a pharyngeal/oropharyngeal cuff and can be a functional airway device when placed in the esophagus or the trachea. The majority of Combitube insertions result in an esophageal position. Ventilation is achieved through the multiorificed lumen when placed in the esophagus, and when placed in the trachea, ventilation occurs via the distal lumen.

Regardless of placement, the distal tracheal tube cuff separates the gastrointestinal system from the respiratory system, and the proximal cuff seals the nasal and oral passages. Since the Combitube is a supraglottic airway, pathology located at or below the level of the larynx may prevent the Combitube from being effective. The Combitube is available in two sizes: 37-French and 41-French.

When inserting the Combitube, the lower jaw and tongue are lifted anteriorly, and the tube is placed with a caudad-curved, downward motion until the two black rings that are printed on the tube are at the level of the teeth. These black rings represent the proximal depth of the tube. The oropharyngeal cuff is inflated with 85-100 cc's of air while the distal balloon is inflated with 5-15 cc's of air. An anesthesia circuit or ambu bag is then attached to the proximal end of the esophageal lumen which is made of blue polyvinyl chloride. If the Combitube has an esophageal placement, ventilation will occur through the lumen's perforations located in the hypopharynx. If breath sounds cannot be auscultated and/or stomach inflation is observed, the Combitube is located in the trachea. The Combitube is not repositioned; however, ventilation is changed to the distal end of the tracheal lumen which is made of clear polyvinyl chloride. If ventilation fails, the Combitube is most likely placed too deeply in the esophagus with the oropharyngeal cuff blocking the airway. Both cuffs should be deflated and the device withdrawn 2 cm.

Combitubes are advantageous in that they provide rapid control of the airway, protect the airway from regurgitation, they are easy to insert, the patient's neck may remain in a neutral position, and visualization of the larynx is not necessary. Contraindications for Combitube use include caustic agent ingestion, mass or foreign body of the upper airway, esophageal obstruction, intact gag reflex, < 4 feet in height, and lower airway obstruction. Combitubes contain latex, therefore, they should not be used in patients who are allergic to latex. Lacerations to the esophageal wall and the pyriform sinus may occur with Combitube use. This can result in pneumomediastinum, subcutaneous emphysema, esophageal rupture, and pneumoperitoneum.

Sandberg W, Urman RD, and Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier; 2011: 88-89.

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